Dale T. Clement, PhD

Curriculum Vitae

Wake Forest University
Department of Biology
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SKILLS

Subject-matter expertise: Quantitative ecology, conservation science, evolution, epidemiology, statistics, applied math

Communication: Scientific writing (8 peer-reviewed publications & 2 preprints), Peer review (8 articles, 1 grant), verbal communication (13 conference presentations), grant writing

Leadership: Project management, experience chairing committees (2 yrs), mentorship (4 mentees), interdisciplinary teamwork (6 projects), international collaboration (6 publications)

Technical Skills: Population viability analysis, demographic modeling, agent-based simulation, maximum likelihood & Bayesian inference, spatial statistics, machine learning, uncertainty analysis

Coding: R (tidyverse, sf, sp, ggplot2), C++, bash scripting, high-performance computing ∗ **Workflow**: Git, quarto

EDUCATION

Ph.D., Population Biology, University of California - Davis Sep 2017 – Aug 2022
Dissertation: Modeling the effects of evolutionary history and stochasticity on population responses to rapid environmental change

B.A., Biology and Mathematics, Dartmouth College Sep 2013 – June 2017
Magna cum laude with High Honors in Biology

EXPERIENCE

Postdoctoral Researcher Sep 2024 – Current
Nicholas Kortessis Lab, Department of Biology, Wake Forest University *Winston-Salem, NC*

Led an interdisciplinary team in developing epidemiological models that incorporate responses of human travel behavior to perceived disease risk, identifying circumstances in which changes in travel behavior increase disease transmission.
- Integrated metapopulation epidemiological models with human risk perception models and risk-dependent travel behavior.
- Created **reproducible model pipelines** with code, metadata, and model outputs deposited on Zenodo.
- Currently integrating models with cell phone mobility and survey data collected by collaborators.

Worked with public health stakeholders (Forsyth County Department of Public Health) to map mosquito and West Nile Virus (WNV) occurrence to identify priority trapping locations to maximize WNV detection.
- Used **spatiotemporal statistics** to analyze patterns of mosquito abundance and West Nile Virus.
- Currently applying **species distribution modeling** to predict environmental factors associated with *Culex spp.* abundance and West Nile Virus prevalence, using land use, climate, eBird, and mosquito trap data.
- Delivered monthly **updates to stakeholders** via in-person meetings and **Quarto-generated reports**.

Postdoctoral Research Associate (40 hr/wk) Sep 2022 – Aug 2024
Andrew Storfer Lab, School of Biological Sciences, Washington State University *Pullman, WA*

Used **agent-based simulations** to conduct **population viability analysis** for the endangered Tasmanian devil (*Sarcophilus harrisii*) and its transmissible tumor.

Project 1: Incorporated quantitative trait evolution for the first time to assess the effect of devil-tumor coevolution on devil persistence, yielding more accurate extinction risk estimates than purely demographic models.

- Created reproducible model pipelines available on Zenodo, with model outputs and metadata available on Dryad following FAIR data principles.

Project 2: Evaluated the impact of conservation intervention on Tasmanian devil populations, including rigorous quantification of model uncertainty, demonstrating that translocation would have only minor effects on long-term devil abundance (manuscript in prep).

- Used machine learning (**XGBoost**) to identify model parameters most strongly influencing devil population persistence and translocation efficacy.
- Used **principal component analysis** and **k-means cluster analysis** to classify simulations into seven discrete model scenarios.

Both Projects:

- Collaborated extensively with Australian experts in devil ecology and evolution to shape model design and interpretation.
- Utilized WSU's Kamiak high-performance computing cluster to run simulations in parallel.
- Validated models with **Approximate Bayesian Computation** using genomic, demographic, and disease prevalence data, confirming that model predictions were consistent with observed population patterns

Contributed to **three grant proposals** and reports:

- Contributed to an NSF Partnership to Advance Conservation Science grant proposal by **assessing the feasibility of a grant aim** that would use genomic offsets to predict the optimal source and recipient populations for translocations conducted on Pacific Northwest amphibian species to facilitate climate adaptation. Determined that insufficient data would be collected to carry out the analysis, leading the aim to be removed from the grant.
- Contributed **preliminary findings** to an NIH grant proposal by using **maximum likelihood estimation** to estimate the growth rates of four tumor lineages from Tasmanian devil **mark-recapture data**.
- Contributed **preliminary findings** to a Washington Research Foundation interim grant report by using **nonlinear mixed models** to compare in-vitro growth rate of two cultured tumor lines in isolation and in competition with each other.

Mentored two graduate students in quantitative methods:

- Assisted a **masters' student** with **mixed effect modeling** and **nonlinear regression** to estimate in-vitro growth rate of cultured tumor lines
- Advised a **PhD student** on using agent-based simulations to test the reliability of estimators of gene-environment associations at different spatial scales.

National Science Foundation Graduate Research Fellow Sep 2017 – Aug 2022
Sebastian Schreiber Lab, Center for Population Biology, University of California, Davis *Davis, CA*

Developed theoretical frameworks to understand the effect of rapid evolution on natural populations' ability to cope with environmental change, yielding critical insights into species' resilience.

- Used evolutionary game theory to model the effects of tree masting on the evolution of seed caching in rodents.
- Used stochastic phenotypic plasticity models to study how rapid environmental change interacted with previously evolved strategies for coping with environmental variability to determine population growth rates.
- Used eco-evolutionary modeling to partition the effects of genetic drift and demographic stochasticity on extinction risk.
- Made **reproducible data pipelines** available on Zenodo and Dryad data repositories.

Served in graduate student leadership for four years:

- **Graduate Student Association representative** (2019-2022): Communicated important campus-wide updates and information to students in my department and voted on policies and resolutions affecting graduate students.
- **Population Biology Student Diversity Committee** (member 2019-2022, chair 2021-2022): Drafted recommendations for departmental policy changes to graduate admissions (adopted), departmental email policy (adopted), & curriculum, compiled resources for future students into an unofficial student handbook, served as Mentor for the 2021 Ecology & Evolution Graduate School Preview Program.
- **Student coordinator** (2020-2021): Coordinated and organized other members of student leadership, mediated between departmental stakeholders, and communicated student concerns to faculty leadership.
- **Other roles**: Peer mentor for incoming graduate students. Served on student panels providing qualifying exam advice.
- Led and participated in scientific outreach, including educational activities at two local middle schools and award-winning, interactive exhibits at the university's open house.

Undergraduate Research Assistant (10 hr/wk) Jun 2015 – Jun 2017
Ryan Calsbeek Lab, Department of Biological Sciences, Dartmouth College *Hanover, NH*

- Collected and photographed wood frogs (*Rana sylvatica*) over two field seasons under graduate student supervision.
- Helped develop protocols for analyzing wood frog coloration from image data.
- Conducted an independent experiment measuring the effect of temperature on coloration in wood frog tadpoles to assess the role of melanization in thermal regulation.

Intern (40 hr/wk) Mar 2015 – Jun 2015
Department of Molecular Ecology, Max Planck Institute for Chemical Ecology *Jena, Germany*

- Screened transgenic lines of coyote tobacco (*Nicotiana attenuata*) to ensure successful expression of the transgene.

PUBLICATIONS

1. R. Zwolak, **D.T. Clement**, A. Sih, S.J. Schreiber. **In Press**. Adaptive shifts in rodent caching behavior buffer plant recruitment amid rapid environmental change. Ecology.

2. **D.T. Clement**, R.D. Holt, N.W. Ruktanonchai, O. Saucedo, N.A. Kortessis. **In Press**. Host behavioral responses to perceived risk shape spatial disease dynamics. Population Biology: Modeling and Theory. Preprint: https://doi.org/10.64898/2026.05.25.726839

3. N.W. Ruktanonchai, N.A. Kortessis, **D.T. Clement**, O. Saucedo, E. Cleary, S. Lai, S. Gao, R.D. Holt. 2026. Accounting for a typology of travel behavior in modeling infectious disease transmission, detection, and community response. Scientific Reports. https://doi.org/10.1038/s41598-026-60383-3

4. **D.T. Clement**, D. Gallinson, R. Hamede, M. Jones, M. Margres, H. McCallum, A. Storfer. 2025. Coevolution promotes the coexistence of Tasmanian devils and a fatal, transmissible cancer. Evolution. qpae143. https://doi.org/10.1093/evolut/qpae143

5. R. Zwolak, **D.T. Clement**, A. Sih, S.J. Schreiber. 2023. Granivore abundance shapes mutualism quality in plant-scatterhoarder interactions. New Phytologist. https://doi.org/10.1111/nph.19443

6. **D.T. Clement**, I.P. Neylan, N.J. Roberts, S.J. Schreiber, P.C. Trimmer, A. Sih. 2023. Evolutionary history mediates population response to rapid environmental change through within-generational and transgenerational plasticity. American Naturalist 201: E90-E109. https://doi.org/10.1086/723624

7. R. Zwolak, **D.T. Clement**, A. Sih, S.J. Schreiber. 2021. Mast seeding promotes evolution of scatter-hoarding. Philosophical Transactions of the Royal Society B. 376: 20200375. https://doi.org/10.1098/rstb.2020.0375

8. D. Goedert, **D.T. Clement**, R. Calsbeek. 2021. Evolutionary trade-offs may interact with physiological constraints to maintain color variation. Ecological Monographs 91: e01430. https://doi.org/10.1002/ecm.1430

SUBMITTED MANUSCRIPTS

1. H. Uecker, M. Osmond, C. Alejandre, E. Barrios-Caro, **D. Clement**, …, M. Yamamichi. **In Review**. Quantifying the Probability of Evolutionary Rescue. Evolution. Preprint: https://doi.org/10.32942/X2TX1R

2. H. Uecker, M. Osmond, C. Alejandre, E. Barrios-Caro, **D. Clement**, …, M. Yamamichi. **In Review**. Modeling Evolutionary Rescue. Evolution. Preprint: https://doi.org/10.32942/X25D4T

GRANTS / FELLOWSHIPS

2019	Graduate Student Association Travel Award	$500
2017-2022	National Science Foundation Graduate Research Fellowship	$102,000
2017-2022	Dean's Distinguished Graduate Fellowship	$27,000
2017	James O. Freedman Presidential Scholar	$2,400
2017	Christopher G. Reed Biologist Award	$500
Summer 2016	Dartmouth Leave Term Grant	$4,500
Fall 2015	Dartmouth Leave Term Grant	$4,500

PRESENTATIONS

Invited Talks

1. **D.T. Clement**. 2026. Host behavioral responses to perceived risk shape spatial disease dynamics. Invited Speaker, MathBio Seminar, Virginia Tech.
2. **D.T. Clement**. 2024. Evolutionary Rescue: Contingency and Uncertainty. Invited Speaker, Ecology, Evolution, and Behavior Seminar Series. Wake Forest University.
3. **D.T. Clement**. 2022. Modeling the effects of evolutionary history and stochasticity on population responses to rapid environmental change. Invited Speaker, School of Biological Sciences Seminar Series, Washington State University.
4. **D.T. Clement**. 2022. Modeling the effects of evolutionary history and stochasticity on population responses to rapid environmental change. Invited Speaker, Center for Population Biology Seminar Series, University of California, Davis.
5. **D.T. Clement**. 2020. When can evolution prevent extinction? Invited Speaker, Center for Population Biology Seminar Series, University of California, Davis.

Contributed Talks

1. **D.T. Clement**, R.D. Holt, N.W. Ruktanonchai, O. Saucedo, N.A. Kortessis. 2026. Does integrating multiple sources of information correct detrimental host behavioral responses to perceived disease risk? Oral Presentation. Ecological Society of America 2026, Salt Lake City, UT.
2. **D.T. Clement**, R.D. Holt, N.W. Ruktanonchai, O. Saucedo, N.A. Kortessis. 2025. Host behavioral responses to perceived risk shape spatial disease dynamics. Oral Presentation. Ecological Society of America 2025, Baltimore, MD.
3. **D.T. Clement**, D. Gallinson, R. Hamede, M. Jones, M. Margres, H. McCallum, A. Storfer. 2024. Rapid coevolution promotes the coexistence of Tasmanian devils and a fatal transmissible cancer. Oral Presentation in Evolutionary Rescue: Theory, Data, and Pressing Applications. Evolution 2024, Montreal, Canada.
4. **D.T. Clement**, D. Gallinson, R. Hamede, M. Jones, M. Margres, H. McCallum, A. Storfer. 2023. Coevolutionary rescue in disease models: Tasmanian devils (Sarcophilus harrisii) as a case study. Oral Presentation, Ecological Society of America 2023, Portland, OR.
5. **D.T. Clement**. 2023. Decomposing the effects of demographic, sex-ratio, and phenotypic stochasticity on extinction risk during evolutionary rescue. Oral Presentation, Mathematical Models of Evolutionary Rescue, Max Planck Institute for Evolutionary Biology, Plön, Germany.
6. **D.T. Clement**. 2023. Can the rapid evolution of disease resistance prevent the extinction of Tasmanian devils (Sarcophilus harrisii)? Oral Presentation, EVO-WIBO, Port Townsend, WA.
7. **D.T. Clement**. 2021. Decomposing the effects on demographic stochasticity and genetic drift on extinction risk during evolutionary rescue. Oral Presentation, Ecological Society of America 2021, Virtual.
8. **D.T. Clement**. 2021. Decomposing the effects on demographic stochasticity and genetic drift on extinction risk during evolutionary rescue. Oral Presentation, Evolution 2021, Virtual.
9. **D.T. Clement**, I.P. Neylan, N.J. Roberts, S.J. Schreiber, P.C. Trimmer, A. Sih. 2021. Evolutionary History, and Effects of Within-Generational and Transgenerational Plasticity on Fitness after Rapid Environmental Change. Oral Presentation, Virtual Asilomar 2021, Virtual.
10. **D.T. Clement**. 2020. Evolutionary rescue in small populations. Oral Presentation, American Society of Naturalists 2020, Pacific Grove, CA.

Posters

1. **D.T. Clement**, R.D. Holt, N.W. Ruktanonchai, O. Saucedo, N.A. Kortessis. 2026. Does integrating multiple sources of information correct detrimental host behavioral responses to perceived disease risk? Poster Presentation. Ecology and Evolution of Infectious Diseases (EEID) 2026, Blacksburg, VA.
2. **D.T. Clement**, R.D. Holt, N.W. Ruktanonchai, O. Saucedo, N.A. Kortessis. 2025. Host behavioral responses to perceived risk shape spatial disease dynamics. Poster Presentation. Ecology and Evolution of Infectious Diseases (EEID) 2025, South Bend, IN.
3. **D.T. Clement**, I.P. Neylan, N.J. Roberts, S.J. Schreiber, A. Sih, P.C. Trimmer. 2019. Transgenerational plasticity mediates fitness loss under rapid environmental change, Poster Presentation, Evolution 2019, Providence, RI.

TEACHING

Teaching Assistantships

2020, Spring	EVE 100- Introduction to Evolution, Department of Evolutionary and Ecology, University of California, Davis (Instructor: Catherine Rushworth)
2020, Winter	BIS 002B- Introduction to Biology: Principles of Ecology & Evolution, Department of Evolution and Ecology, University of California, Davis (Instructor: Jay Rosenheim)

Specialized Training

2021, Fall	EVE 298: Instructional Design of Large Series Classes (Instructor: Susan Keen)
2020, Fall	ACCELERATE Asynchronous (Self-directed course on online teaching, taken as part of an informal departmental reading group)
2020, Winter	EVE 298: Biology Education Research & Evidence-Based Teaching (Instructor: Laci-Gerhart Barley)

LEADERSHIP, SERVICE, & OUTREACH

Grant Reviews

2026	**Outside Reviewer**, European Research Council
Journal Reviews	Ecology, Ecology Letters, Ecological Modeling, Evolution, Journal of Evolutionary Biology, Royal Society Interface

Leadership and Outreach

2021-2022	**Chair**, Population Biology Student Diversity Committee, UC Davis
2020-2021	**Student Coordinator**, Population Biology Graduate Group, UC Davis
2018-2022	**Member**, Population Biology Student Diversity Committee, UC Davis.
2018-2022	**Graduate Student Association Representative** for Population Biology, UC Davis Graduate Student Association, UC Davis.
2018-2019	**Volunteer**, STEM Squad, Middle School Outreach.